

Ana Clara Loureiro Müller Zaidan

Intelligent Platform for Social Engagement

SÃO PAULO
2025

Ana Clara Loureiro Müller Zaidan

Intelligent Platform for Social Engagement

Final Course Project submitted to the
Institute of Technology and Leadership
(INTELI), to obtain a bachelor's degree in
Software Engineering.

Advisor: Prof. José Romualdo

SÃO PAULO
2025

(Catalog Card (mandatory item – NBR14724, item 4.2.1.1.2))

Cataloging in Publication
Library and Documentation Service
Institute of Technology and Leadership (INTELLI)
Data entered by the author.

(Cataloging record with international cataloging data, according to NBR 14724. The record will be completed later, after approval and before the final version is deposited. The completion of the cataloging record is the responsibility of the institution's library.)

Sobrenome, Nome

Título do trabalho: subtítulo / Nome Sobrenome do autor; Nome e
Sobrenome do orientador. – São Paulo, 2025.

nº de páginas : il.

Trabalho de Conclusão de Curso (Graduação) – Curso de [Ciência da
Computação] [Engenharia de Software] [Engenharia de Hardware] [Sistema
de Informação] / Instituto de Tecnologia e Liderança.

Bibliografia

1. [Assunto A]. 2. [Assunto B]. 3. [Assunto C].

CDD. 23. ed.

Acknowledgments

I would like to express my sincere gratitude to my advisor, Prof. José Romualdo, for his guidance, continuous support, and helpful discussions throughout the entire duration of this project. His expertise was crucial to the successful completion of the "Intelligent Platform for Social Engagement". I also extend my heartfelt thanks to my family for their support, encouragement, and belief in me during my academic journey.

Abstract

[Zaidan, Ana Clara. **Intelligent Platform for Social Engagement**. 2025. 19 pages. Final course project (Bachelor) – Institute of Technology and Leadership, São Paulo, 2025.

This project, "Intelligent Platform for Social Engagement," addresses the critical challenge faced by social organizations, such as the partner "Olhar de Bia," in effectively connecting individuals willing to volunteer with required collaboration opportunities. The urgency of this object of study is highlighted by research indicating that 48.6% of potential volunteers in São Paulo struggle to find suitable projects (CHANG, 2022). The main objective was to develop a centralized web platform that leverages Artificial Intelligence (AI) to suggest events and donation campaigns, ensuring alignment between volunteer profiles and organizational needs.

The methodology followed agile principles, structuring development into four ten-week modules utilizing the PDCA cycle for continuous improvement, and incorporated strategic artifacts such as Customer Analysis using the Onna framework, and process modeling via BPMN.

The proposed solution is a three-layer architecture, modeled with the C4 technique, comprising a React/TypeScript Single-Page Application (SPA) frontend, a Supabase/PostgreSQL backend, and an AI Recommendation Service built with Python. Technical foundations included the definition of Conceptual and Logical Data Models, based on relational database principles (ACID properties and normalization), and environment standardization using Dev Containers and GitHub Codespaces.

Key progress included user research (42 responses) and an Exploratory Data Analysis (EDA) that confirmed a strong user preference for value alignment (mean score 4.8) and logistical constraints, which informed the collaborative filtering recommendation model's item space and filtering logic. The project successfully achieved its foundational goals, providing a robust, data-driven structure for the final implementation of the core AI matching system.

Key-words: [intelligent platform]; [social engagement]; [volunteering]; [recommendation system]; [software architecture].

List of Tables

Table 1 – [User Stories].....	page 12]
-------------------------------	----------

Summary

1 Introduction	8
2 Solution Development	9
2.1 Applied Rationale	9
2.3 Assessment of Impact and Contribution to the Business	16
3 Conclusion	17
References	19

1 Introduction

1.1. Partner Company Context:

The project partner is the social organization "Olhar de Bia," whose mission centers on ending poverty through education. Ms. Beatriz Martins, the NGO's founder, served as the Product Owner and focal point for the project. The NGO acts as a representative of Civil Society Organizations (CSOs) that require collaboration.

The project addresses the strategic need to improve the "fit" between volunteers and an organization's needs/causes and increase the reach of potential volunteers and donors for organizations like Olhar de Bia. From a social impact perspective, the platform aims to stimulate participation in diverse social causes and indirectly guide resources toward organizations.

1.2. Problem Definition (Corporate Pain Point):

The core problem is the difficulty in effectively connecting individuals willing to volunteer with social organizations needing collaboration. Many potential volunteers abandon the process because they do not know how to find projects or are unaware of existing institutions. Conversely, CSOs face challenges in publicizing opportunities and selecting suitable volunteers. Research conducted in São Paulo (CHANG, 2022) indicated that 48.6% of respondents had never participated in volunteer work, citing difficulty finding projects and lack of knowledge about institutions.

Current Metrics (Baseline): The baseline challenge relates to the low participation rate in volunteer work (48.6% of respondents citing non-participation) and the existing friction points in recruitment, such as Raquel's frustration in searching for opportunities and Agnes's insecurity when recruiting reliable volunteers.

1.3. Proposed Solution and Expected Contribution:

The proposed computational solution is the Intelligent Platform for Social Engagement, a web platform that centralizes opportunities and donation campaigns. It utilizes Artificial Intelligence (AI) to connect volunteers with causes and events best aligned with their profiles and interests. The expected results include improving the alignment ("fit") between volunteers and organizations.

1.4. Business Objectives:

The expected results for the partner organization are to increase the reach of potential volunteers and donors, and improve the quality of the match between volunteers and organizational needs. Socially, the project seeks to stimulate participation in various social causes

1.5. Structure of the thesis/dissertation:

This document details the project phases, starting with the rationale for technological and methodological choices, followed by the technical specifications, architecture, and implementation details,. Finally, it outlines the metrics and planned assessment of impact, followed by the conclusions and references

2 Solution Development

2.1 Applied Rationale

2.1.1 Business Area Rationale:

Currently, there are products in the market aimed at publicizing NGO events and connecting them with volunteers. However, a pain point is the lack of personalized matching between events and people, generating superficial connections that do not add long-term value for the NGOs.

This scenario reflects the challenge in effectively connecting individuals willing to volunteer with social organizations, leading to difficulties for CSOs in selecting suitable volunteers. The project specifically addresses this by developing a solution that uses Artificial Intelligence to connect volunteers with causes and events best aligned with their profiles and interests, intending to improve the "fit" between volunteers and the organization's needs/causes.

The project began with an in-depth Customer Analysis to ensure the platform meets real user needs. This analysis utilized the Onna framework for data-driven persona creation, combining qualitative and quantitative methodologies. The Customer Journey Mapping was based on the Nielsen Norman Group model, focusing on expectations, emotions, and pain points. The key pain points identified, such as difficulty finding suitable opportunities and insecurity in recruiting, were crucial in identifying platform improvement opportunities, such as the effective recommendation system and validation tools.

2.1.2 Technological rationale for the solution:

The platform is designed with a three-main-layer structure (frontend, backend API, and lightweight recommendation service).

The AI-Based Recommendation Service uses Python, Flask (for API exposure), Pandas, and scikit-learn for machine learning algorithms, alongside spaCy for natural language processing (to interpret user interests).

The C4 model (developed by Simon Brown) was used to model the architecture, and to describe the solution architecture across multiple levels of abstraction (Context and Containers), facilitating clear communication among stakeholders.

Considering data persistence, the choice of PostgreSQL as the relational database is justified by the theoretical foundations of relational databases, including ACID

properties, data integrity and consistency, security, and the use of the normalization process to optimize data structures.

Dev Containers were adopted to standardize and isolate the development environment, packaging dependencies and configurations, and GitHub Codespaces utilizes Dev Containers to provision ready-to-use cloud-based environments.

The technology stack was chosen considering the available time and monetary resources, prioritizing robustness, community support and the developer's familiarity or interest in learning the tools. The stack includes React.js and TypeScript for dynamic, scalable SPAs (Frontend), Supabase for Backend-as-a-Service, authentication, and database management, and the Python ecosystem (Flask, Pandas, scikit-learn) for the personalized recommendations.

2.1.3 Fundamentals of Management and Development Methods:

Project management follows agile principles adapted to the academic context. The development is organized into two-week sprints within four ten-week modules. The project utilizes a continuous improvement cycle based on the PDCA (Plan, Do, Check, Act) model, aligned with Scrum rituals for reviewing deliverables and implementing process improvements each sprint. BPMN (Business Process Model and Notation) was utilized to map essential workflows (e.g., "Company Registers an Event," "User Registers for an Event"), standardizing communication and identifying improvement areas.

2.2 Specification and Development :

2.2.1 Requirements and Specifications:

The core **features** of the project are defined as follows:

Organization Management

- Registration of organizations;
- Registration and authentication of organizations' representatives;

- Editing organizational information (name, description, contact details, and social media links).

Event Management

- Creation and publication of events by organizations, including event date, location, description, and available volunteer slots;
- Editing and deletion of published events;
- Volunteer registration for events.

Volunteer Management

- Registration and authentication of volunteers;
- Volunteer profile management, including information on interests and prior experiences;
- Display of available events for volunteer engagement.

Main Feed / Explore Section

- Personalized event recommendations based on the user's profile;
- Advanced filtering for events and organizations;
- Search functionalities based on event name and the responsible organization.

At a broader level, aiming to capture the users' intentions when interacting with the platform, the following **User Stories** have been defined:

ID	User Story
US01	As an organization representative, I would like to register my NGO on the platform, to make it visible to potential volunteers and donors.
US02	As an organization representative, I would like to register and log in on the platform, to manage my NGO's activities securely.
US03	As an organization representative, I would like to create and publish events, to attract volunteers to participate in our initiatives.
US04	As an organization representative, I would like to edit and/or delete events, to maintain accurate and updated information about volunteer opportunities.
US05	As an organization representative, I would like to edit my NGO's information, to ensure the profile remains current and aligned with our mission.
US06	As a volunteer, I would like to register and log in on the platform, to access and manage my participation in volunteer opportunities.
US07	As a volunteer, I would like to receive personalized event recommendations based on my interests, to find opportunities that match my profile and preferences.

ID	User Story
US08	As a volunteer, I would like to search for events using keywords and filters, to find opportunities relevant to my availability and interests easily.
US09	As a volunteer, I would like to apply for events organized by NGOs, to engage in causes that resonate with me.
US10	As a volunteer, I would like to manage my profile, such as editing my interests and updating personal information, to ensure my data and preferences remain accurate.
US11	As a volunteer, I would like to view all events I have applied for, with filters to distinguish between past and upcoming events, to better organize my volunteering activities.

Table 1 - User Stories
Source: *Internally Developed*

Based on these definitions, specific tasks were established to ensure that each user story was adequately addressed, forming the product backlog to be distributed across the sprints of each development module.

2.2.2 Architecture and Technology:

To effectively describe the solution architecture of our system, we will utilize the C4 model - a framework designed to visualize software architecture across multiple levels of abstraction. This model facilitates clear communication among stakeholders by providing a structured approach to representing complex systems.

The C4 model, developed by software architect Simon Brown, comprises four hierarchical levels:

- **Context:** Offers a high-level overview of the system, illustrating its interactions with users and external systems.
- **Containers:** Breaks down the system into containers such as applications and databases, detailing their responsibilities and interactions.
- **Components:** Delves into the internal structure of each container, highlighting the components and their relationships.
- **Code:** Provides detailed views of the source code, often omitted in high-level architectural discussions due to its granularity.

In this document, we will focus on the first two levels - Context and Containers - to provide a comprehensive yet accessible depiction of the system's architecture. This approach ensures that both technical and non-technical stakeholders can understand the system's structure and interactions without being overwhelmed by intricate details.

The C4 model has been recognized for its effectiveness in enhancing communication and understanding within software development teams. As noted by Brown, "The C4 model is an easy to learn, developer friendly approach to software architecture diagramming" (BROWN, s.d.). By adopting this model, we aim to create clear and consistent architectural diagrams that facilitate collaboration and decision-making throughout the development process

Level 1: Context Diagram

The Context Diagram presents a high-level overview of how the *Intelligent Platform for Social Engagement* interacts with its primary users and external systems. It highlights the relationships between the platform, volunteers, non-governmental organizations (NGOs), and supporting services. This diagram is particularly valuable for stakeholders - including the development team and representatives from *Olhar de Bia* - as it clarifies the system's purpose and scope without delving into implementation details.

The platform serves as a bridge between volunteers and social organizations (NGOs), fostering engagement in social initiatives. Volunteers interact with the platform to discover and participate in meaningful projects, while NGOs use it to publish their initiatives and connect with potential contributors. To enhance the user experience, an AI Recommendation Service delivers personalized event suggestions based on individual interests.

Level 2: Container Diagram

The Container Diagram offers a deeper view of the platform's internal structure by detailing the main containers (applications, services, and databases) that make up the system. It illustrates how responsibilities are distributed across different parts of

the architecture and identifies the key technologies in use. It supports decisions related to development, integration, and deployment.

The system is composed of several key components that work together to deliver a seamless experience. The Web Application frontend is built with TypeScript and Angular, which serves as the primary user interface for interacting with the platform. Supporting this interface is the API Server, developed using Supabase, a backend as a service platform which manages the business logic and handles all data processing operations. For persistent data storage, the platform relies on a PostgreSQL database. Enhancing user engagement, the AI Recommendation Service, built with Python and Flask, provides personalized suggestions based on user interests.

2.2.3 Development and Implementation (MVP):

The implementation utilized React with TypeScript and SASS for styling, following the Figma design. Authentication and session management are integrated via Supabase and Google OAuth.

Regarding data collection & EDA, a User research was conducted via an online questionnaire (42 responses), which covered demographics, barriers to engagement, value alignment, areas of interest, and logistical preferences. An Exploratory Data Analysis (EDA) was performed using Python (pandas, NumPy, Matplotlib), revealing key findings such as moderate difficulty finding opportunities (3.3/5) and strong preference for value alignment (4.8 mean score). These findings were used to construct datasets and define the structure for the collaborative filtering recommendation model.

The current implementation of the solution includes a functional web application that implements all core functionalities as described. It provides navigation across core pages (home, login, profile, events) and establishes the visual identity, structural foundation, and tested authentication flow.

The pilot deployment process utilizes Vercel for the frontend, and AWS (Amazon Web Services) services for the backend, throughout Supabase Free Trial Plan. The Recommendation API runs in an Azure virtual machine.

2.2.4 Testing and Technical Evaluation:

The technical evaluation criteria are defined by the Success Metrics:

- Availability: Target of > 99.5% monthly uptime.
- Mean Time to Recovery (MTTR): Target of < 2 hours average time to restore the system after failure.
- Average Latency: Target of < 300ms API response time.
- Code Coverage: Target of > 70% of code covered by automated tests.

2.3 Assessment of Impact and Contribution to the Business

2.3.1 Defining Corporate Success Metrics:

Project KPIs (Key Performance Indicators) were defined based on three key pillars:

- Business: Active users, active institutions, volume of event and project publications.
- Technology: Number of accesses, resource usage (CPU, memory), number of bugs identified/resolved, security incidents.
- Project Management: Completed vs. planned deliveries, delivery delays, rework, developer productivity. Measurement Methodology: Risk and change management are handled through dynamic backlog prioritization and continuous stakeholder feedback. Communication is maintained via weekly and bi-weekly meetings with the advisor (Prof. José Romualdo), utilizing Trello and GitHub for transparency.

2.3.2 Results and Impact Analysis:

Quantitative Analysis: The expected quantitative results are tied to the Success Metrics, aiming for high platform reliability (e.g., Availability > 99.5%) and successful user engagement (User Retention > 50%).

Qualitative Analysis: The qualitative gain is improved user/internal customer satisfaction through an intuitive UI/UX and increased agility in connecting volunteers with opportunities based on strong value alignment.

2.3.3 Cost-Benefit Analysis:

- Estimated development/implementation costs (labor, licenses, infrastructure) ;
- Calculation of **Return on Investment (ROI)** or other financial indicators relevant to the company (e.g. , *Payback*).

2.3.4 Critical Success Factors and Lessons Learned:

Critical factors for success include continuous feedback and backlog prioritization,, the use of standardized frameworks like C4 and BPMN,, and grounding UX decisions in scientific literature. Lessons learned frequently involved the importance of time management and realistic scoping,, the value of continuously updated backlogs, and the process of translating analytical outcomes (EDA) into practical design recommendations for the AI system.

3 Conclusion

The project successfully achieved its foundational objectives by defining a robust architecture, establishing a comprehensive development environment, conducting essential user research and data analysis, and developing a functional web application. The problem of volunteer/organization misalignment was thoroughly researched and translated into actionable technical specifications for the AI Recommendation System,. All planned artifacts for the initial modules, including Customer Analysis, Process Modeling, Design, Architecture, Data Modeling, and

Environment Setup, were completed, leveraging continuous improvement cycles (PDCA) to address challenges such as scope management and technical knowledge gaps.

Regarding the Knowledge Transfer Plan, technical documentation and the code repository are centralized on GitHub, structured into key modules, with respective internal documentation and instructions.

References

CHANG, Fernanda Li. **Volunteer Match:** connection between civil society organizations and volunteers. Limeira, 2022.